Name: ______________________

TOPIC TEST

Equations and inequalities (Advanced)

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

1 Solve $x - 11 = 2(5 - 3x)$.

A $x = 3$

B $x = -\frac{21}{5}$

C $x = -\frac{1}{5}$

D $x = -\frac{1}{7}$

2 Solve $3n + 5 \leq n + 1$.

A $n \leq -4$

B $n \leq -2$

C $n \geq -2$

D $n \geq 4$

3 Evaluate $4 - |-3|^2$.

A 13

B 10

C –2

D –5

4 Solve $|2y - 3| = 9$.

A $y = -3$

B $y = 6$

C $y = -3, y = 6$

D $y = -6, y = 3$

5 Solve $2a^3 + 11 = -5$.

A $a = -8$

B $a = -2$

C $a = 2$

D $a = 8$

6 Solve $m^2 - 2m = 15$.

A $m = -3$ or -5

B $m = -3$ or 5

C $m = 3$ or -5

D $m = 3$ or 5

7 Solve $2k^2 - 5k - 1 = 0$.

A $k = \frac{5 \pm \sqrt{33}}{4}$

B $k = \frac{-5 \pm \sqrt{33}}{4}$

C $k = \frac{5 \pm \sqrt{17}}{4}$

D $k = \frac{-5 \pm \sqrt{17}}{4}$

8 The formula to calculate the volume of a cone is $V = \frac{1}{3}\pi r^2 h$.

Find the value of h when $V = 95.3$ and $r = 2.4$.

A 49.6

B 37.9

C 15.8

D 1.8

9 Solve simultaneously $2x - y = 2$ and $x + 4y = 1$.

A $x = 1, y = 1$

B $x = -1, y = 0$

C $x = 1, y = 0$

D $x = 0, y = 1$

10 Solve $\frac{2}{2x - 1} = \frac{4}{x}$.

A $x = \frac{2}{3}$

B $x = \frac{3}{2}$

C $x = -\frac{3}{2}$

D $x = -\frac{2}{3}$

Part B

10 free-response questions
40 marks
Show your working where appropriate.

11 Solve each equation.

a $1-\frac{2p}{3} = 5$

b $\frac{x+5}{2}+\frac{x-2}{3} = 1$

[6 marks]

12 Solve $3(1 - x) \le 2x + 5$ and graph the solution on a number line.

[4 marks]

13 Solve $|5w - 3| = 2$.

[4 marks]

14 Show that $|a + b| \leq |a| + |b|$ when $a = 2$ and $b = -5$.

[3 marks]

15 Solve $3^{x-1} = \frac{1}{9}$.

[3 marks]

16 Solve $(2x + 5)(x + 1)(x - 4) = 0$.

[3 marks]

17 Solve each equation.

a $(y + 5)^2 = 4$

b $3n^2 + 2n = 1$

[6 marks]

18 Given $s = ut + \frac{1}{2}at^2$, find the value of a if $s = 30.7404$, $u = 5.2$, $t = 3.6$.

[3 marks]

19 Solve simultaneously $y = x - 5$ and $y = x^2 + 6x - 1$.

[4 marks]

20 Solve these simultaneous equations.

$A + B = 2$

$A + B + 2C = 1$

$2A - B + 4C = 8$

[4 marks]

This is the end of the test.

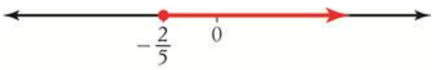
Answers

1 A **2** B **3** D **4** C **5** B

6 B **7** A **8** C **9** C **10** A

11 **a** $p = -6$ **b** $x = -1$

12 $x \geq -\frac{2}{5}$



13 $w = \frac{1}{5}$ or 1

14 LHS = $|2 + -5|$
$= |-3|$
$= 3$
RHS = $|2| + |-5|$
$= 2 + 5$
$= 7$
$\therefore 3 < 7$

15 $x = -1$

16 $x = -\frac{5}{2}, \ -1, 4$

17 **a** $y = -7, -3$ **b** $n = -1, \frac{1}{3}$

18 $a = 1.855$

19 $x = -1, y = -6$ and $x = -4, y = -9$

20 $A = 4, B = -2, C = -\frac{1}{2}$